

Executive Summary – Telecom Customer Churn Analysis

This exploratory data analysis (EDA) examines customer churn behavior in a telecom company dataset to identify the most influential drivers of customer attrition and uncover opportunities to improve retention. The analysis combines demographic, contractual, billing, and service-related factors with visual insights from multiple charts and distribution analyses.

Overall Churn Snapshot

- The dataset contains 7,043 customer records.
- 1,869 customers churned, resulting in an overall churn rate of 26.54%.
- 73.46% of customers remained with the company, indicating a solid retention base but a meaningful churn problem that directly impacts recurring revenue.

Key Insights from the Analysis

1. Gender Is Not a Significant Churn Driver

- The customer base is almost evenly split by gender.
- Churn rates among male and female customers are very similar, with only a marginal difference.

Business implication:

Gender-based retention campaigns are unlikely to deliver meaningful impact. Retention strategies should instead focus on behavioral and contractual factors.

2. Senior Citizens Are More Likely to Churn

- Senior citizens make up a smaller share of the customer base, but their churn rate is noticeably higher than that of non-senior customers.
- Non-senior customers account for the majority of retained users, while senior customers are disproportionately represented among churned users.

Business implication:

Older customers may require simpler plans, better customer support, and clearer communication around billing and services.

3. Early Customer Tenure Is the Highest-Risk Period

- Customers with very short tenure (especially 1–2 months) show the highest churn concentration.
- Churn declines steadily as tenure increases, and long-term customers are far more loyal.

- The median tenure among churned customers is substantially lower than among retained customers.

Business implication:

The onboarding period is critical. Proactive engagement during the first 90 days can significantly reduce churn.

4. Contract Type Strongly Influences Retention

- Month-to-month contracts have the highest churn rate and dominate the churned customer segment.
- One-year contracts show much lower churn.
- Two-year contracts have the strongest retention performance and the lowest churn rate.

Business implication:

Longer contract commitments are highly effective in stabilizing the customer base. Incentivizing upgrades from month-to-month plans could materially reduce churn.

5. Value-Added Services Reduce Churn

- Customers subscribed to additional services such as Online Security, Device Protection, and Tech Support are less likely to churn.
- Customers without these services churn at a higher rate, suggesting lower engagement and perceived value.

Business implication:

Bundling complementary services increases customer stickiness and perceived value, making customers less likely to switch providers.

6. Payment Method Matters

- Customers using Electronic Check have the highest churn tendency among payment methods.
- Automated payment options (such as bank transfer or credit card auto-pay) are associated with better retention.

Business implication:

Encouraging customers to move to automated payment methods may improve retention and reduce friction in the billing experience.

7. Monthly Charges and Total Charges Patterns

- Churned customers tend to have higher monthly charges on average than retained customers.
- However, retained customers often have higher total charges because they stay with the company longer.

Business implication:

High monthly pricing without strong perceived value may contribute to churn, while long-term customers naturally accumulate greater lifetime value.

Visual Analysis Highlights

- Count plots clearly illustrate the imbalance between retained and churned customers and highlight the dominance of month-to-month contracts among churned users.
- Boxplots for tenure and charges reveal strong separation between churned and retained customers, especially for tenure.
- Pie charts and percentage distributions make demographic and service adoption patterns easy to interpret for business stakeholders.
- Stacked and grouped bar charts effectively compare churn behavior across contract types, payment methods, and service categories.

Strategic Recommendations

Short-Term Actions (0–3 months)

1. Improve onboarding for new customers
Focus on the first 90 days with welcome campaigns, usage guidance, and proactive support outreach.
2. Target high-risk month-to-month customers
Use churn propensity indicators (short tenure, high monthly charges, electronic check payments) to trigger retention offers.
3. Promote auto-pay adoption
Offer incentives for customers to switch from electronic checks to automated payment methods.

Medium-Term Actions (3–9 months)

1. Encourage contract upgrades
Provide discounts or bundled benefits for moving from month-to-month to one-year or two-year plans.
2. Expand service bundling
Package Online Security, Tech Support, and Device Protection together to increase perceived value and engagement.

3. Develop senior-focused retention programs
Simplify plan options, improve accessibility, and offer dedicated support channels for senior customers.

Long-Term Actions (9+ months)

1. Build a predictive churn model
Use tenure, contract type, monthly charges, payment method, and service adoption as core predictive features.
2. Measure customer lifetime value (CLV)
Prioritize retention investments on customers with the highest long-term revenue potential.

Conclusion

The analysis shows that customer churn is driven primarily by contract type, tenure, service adoption, senior citizen status, monthly charges, and payment method. The most vulnerable segment consists of customers with short tenure, month-to-month contracts, limited service usage, higher monthly charges, and electronic check payments.

By strengthening onboarding, incentivizing long-term contracts, expanding bundled services, and targeting high-risk customers with proactive retention efforts, the company can meaningfully reduce churn and improve customer lifetime value.